

Cloud Cost Optimization and FinOps Automation

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Abstract. Cloud Computing has become an essential technology for modern organizations, offering flexibility, scalability, and easier access to computing resources. However, the increasing adoption of cloud services has also led to growing operational costs, making cost management a significant challenge for many companies. In this context, FinOps has emerged as a framework that promotes collaboration between technical and financial teams to improve cost visibility and optimize resource utilization. This report examines the main challenges associated with cloud costs, presents the fundamental principles of FinOps, and explores cost optimization techniques and automation strategies. The analysis highlights how the implementation of FinOps practices can help organizations improve financial control, reduce waste, and maximize the value obtained from cloud investments.

Keywords: Cloud Computing · FinOps · Cost Management.

1 Introduction

Cloud Computing has become increasingly important for organizations, providing flexible access to computing resources through the Internet. This technology offers several advantages, such as scalability, reduced investment in physical infrastructure, and greater adaptability to business needs. As a result, its adoption has grown significantly worldwide [13, 12].

However, the increasing use of cloud services has been accompanied by rising costs. Lack of monitoring, inefficient resource usage, and the complexity of cloud environments can lead to unnecessary expenses and make budget control more difficult for organizations [9, 11].

In this context, FinOps has emerged as an approach that aims to improve cloud financial management by promoting collaboration between technical and financial teams. Therefore, this report aims to analyze the main challenges related to cloud costs, present the fundamentals of FinOps, and explore optimization and automation strategies that enable a more efficient use of cloud resources.

2 Cloud Computing Cost Challenges

Cloud computing offers countless advantages, and its growth has become exponential. However, one of its biggest downsides remains cost. Even with the rise

of on-demand services that allow users to pay only for what they consume, it is still common for businesses to waste significant amounts of money on unnecessary cloud resources. This occurs because, although cloud systems are useful and cost-effective when properly managed, the sheer scale of resources involved makes it very easy for costs to spiral out of control without adequate monitoring.

2.1 Common Causes of Cloud Waste

A clear example of this problem is **idle resources**. These include virtual machines, databases, or containers that continue running despite having very low activity or no longer serving any purpose. In many cases, resources used in testing and production environments remain active indefinitely, generating costs without producing value.

Another common issue is **unused storage**. Over time, organizations accumulate large volumes of data that are no longer relevant to the current state of the business. This may include old logs, redundant backups, or outdated snapshots. Since most cloud services charge based on storage usage, unnecessary data directly translates into unnecessary costs.

In addition, **data transfer costs** are often overlooked by organizations. These costs can become a significant portion of overall expenses if not properly managed. This is because unused or unnecessary data is frequently transferred between services, generating additional and avoidable charges.

2.2 Financial Impact

Although the idea of losing money may initially seem straightforward, the consequences are often more complex. One of the most significant impacts is **budget overruns**, which occur when a project exceeds its planned budget. In many cases, this is not a minor deviation; studies show that, on average, budget overruns exceed original estimates by 27%, and one in six projects exceeds the budget by more than 200% [7].

Another major issue is the **lack of visibility**. Since cloud resources are commonly used across teams and workflows, it is easy for many users to create and manage resources independently. This makes it difficult to determine who is responsible for specific costs. Without proper identification mechanisms, cost reporting and monitoring become ineffective, making it nearly impossible to identify optimization opportunities.

All of these issues, combined with the unpredictable nature of cloud costs—due to the dynamic scaling of most services based on usage—place a significant financial burden on organizations. As a result, understanding cloud waste fundamentals and adopting FinOps methodologies is no longer just an optimization strategy, but a critical requirement in modern IT cost management.

3 FinOps Fundamentals

3.1 What is FinOps?

FinOps is a financial management practice applied to cloud services that combines the areas of finance, operations, and technology [8]. Its main objective is to help organizations control, monitor, and optimize cloud costs, ensuring efficient resource utilization and a greater return on investment.

This approach promotes collaboration between technical, financial, and business teams, enabling a better understanding of expenses and more informed decision-making. Therefore, FinOps is not limited to cost reduction; it also seeks to maximize the value obtained from the use of cloud services.

3.2 FinOps Principles

FinOps is based on a set of principles that help organizations manage cloud costs more efficiently. One of the fundamental principles is **collaboration between teams**, involving technology, finance, and business departments in decision-making related to the use of cloud resources [6, 8].

Another important principle is **shared responsibility**, as all teams should be aware of the financial impact of their choices. FinOps also promotes decision-making based on business value, seeking to balance costs, performance, and organizational objectives.

In addition, management should rely on **accurate and up-to-date cost data**, providing greater visibility into cloud spending. The existence of a central coordination function to establish best practices and the focus on **continuous optimization** of resources are also essential aspects of this model.

3.3 FinOps Lifecycle

FinOps is based on a continuous cycle composed of three phases: **Inform, Optimize and Operate**. This cycle enables organizations to continuously monitor, analyze and improve cloud cost management.

During the **Inform** phase, usage and cost data are collected and analyzed, providing greater visibility into cloud spending. The **Optimize** phase focuses on identifying opportunities to reduce costs, such as rightsizing resources or removing unused resources. Finally, in the **Operate** phase, optimizations are implemented and monitored, integrating FinOps practices into the organization's daily operations.

As this is a continuous process, the results obtained are regularly reviewed to identify new improvement opportunities, ensuring efficient cloud cost management over time [5].

4 Cloud Cost Optimization Techniques

Cloud cost optimization techniques are a set of methods and strategies designed to eliminate unnecessary costs and reduce required spending, while maintaining system performance, reliability, and scalability. Since cloud resources typically follow a pay-as-you-go model, it is easy for organizations to overspend. To mitigate this, a range of methods and techniques has been developed.

4.1 Rightsizing Resources

Rightsizing resources refers to adjusting the capacity and configuration of a resource to match its actual requirements. To ensure desired performance, developers often over-provision resources such as virtual machines or containers. While this is not necessarily incorrect at an initial stage, resource specifications must be regularly reviewed to avoid unnecessary costs from underutilized capacity.

By analyzing metrics and logs such as CPU utilization, memory consumption, and network activity, organizations can scale components up or down to achieve an optimal balance between performance and cost efficiency.

4.2 Resource Scheduling

This method involves automatically starting and stopping resources that do not need to run continuously. Many development, testing, and internal systems do not require 24/7 availability, especially outside working hours. To avoid unnecessary costs, this strategy deactivates such resources when they are not needed and reactivates them only when required.

For example, by implementing automated shutdown policies for internal applications, organizations can ensure that resources run only on weekdays and only during necessary hours. This can reduce costs by more than 50% for certain workloads while maintaining equivalent functionality. Applying this approach across multiple systems can lead to substantial overall savings.

4.3 Reserved and Spot Instances

Cloud providers offer multiple pricing models, but some have introduced alternative options to address overspending, which are less commonly used but highly effective in the right contexts.

One such option is **reserved instances**, which allow organizations to commit to a fixed amount of cloud resources over a longer period in exchange for significantly reduced pricing compared to on-demand rates. This model is well suited for workloads with predictable and stable resource requirements.

On the other hand, **spot instances** serve a different purpose. They provide access to unused cloud capacity at a much lower cost, but with the risk that resources may be reclaimed by the provider at any time with limited notice. These instances are ideal for flexible, fault-tolerant workloads that can tolerate interruptions.

4.4 Storage Optimization

Storage optimization focuses on reducing the costs associated with data storage and related operations by allowing organizations to determine how and where their data is stored. Most cloud providers offer multiple storage options, typically organized into different tiers based on access frequency, retrieval requirements, and performance characteristics. As a general rule, frequently accessed data is stored in higher-performance tiers, while infrequently accessed data is moved to lower-cost storage tiers.

Additionally, organizations often accumulate large amounts of irrelevant or outdated data over time. By implementing data lifecycle policies, it becomes possible to automatically move data between storage tiers according to usage patterns and remove data that is no longer needed. This approach reduces storage costs while ensuring that critical data remains readily accessible.

4.5 Tagging and Governance

Tagging and governance is a strategy that involves assigning metadata labels to cloud resources in order to improve organization, cost tracking, and accountability. These tags typically contain information such as the associated project, resource owner, department, or deployment environment.

Tagging enables organizations to accurately track resource usage and allocate costs to the appropriate teams or projects. When combined with governance policies, it is possible to enforce tagging requirements and prevent the creation of resources that do not comply with predefined standards. As a result, organizations can achieve highly accurate cost reporting, improved resource visibility, and greater accountability across cloud environments.

5 FinOps Automation and Case Study

5.1 Role of Automation

Automation plays an important role in FinOps, as it enables cloud costs to be controlled and optimized more efficiently [8]. As the use of cloud services increases, it becomes difficult to perform all monitoring and analysis tasks manually. Therefore, automation helps reduce operational effort and improve the financial management of cloud infrastructure.

One of its main functions is the continuous monitoring of resources, identifying excessive consumption, underutilized resources, and potential waste [1]. In addition, it can generate automatic alerts when specific cost or usage thresholds are exceeded, allowing teams to respond quickly to issues.

Automation also supports resource optimization through recommendations such as rightsizing virtual machines or removing unused resources [4]. In some cases, these actions can even be performed automatically, contributing to more effective cost reduction.

Despite its benefits, automation does not replace human involvement. Teams remain responsible for defining strategies and validating important decisions. In this way, automation acts as a support mechanism for FinOps, enabling more efficient and continuous cloud cost management.

5.2 Case Study

The application of FinOps practices has shown significant results in reducing cloud costs, as demonstrated by several real-world cases.

In one example, a large insurance company with high cloud spending faced issues such as underutilized resources and a lack of cost control. Through measures like rightsizing, elimination of unused resources, and improved governance, the company identified potential savings of several million dollars per year, with a significant portion achievable in the short term [2].

Another case involving a software company showed resource waste due to rapid infrastructure growth. By applying FinOps practices such as removing idle resources and optimizing storage usage, the company was able to significantly reduce operational costs and improve cloud financial management [3].

Similarly, a study by Parsimo shows that an organization managed to reduce around 36% of its AWS costs by improving monitoring, increasing Savings Plans coverage, and eliminating unused resources [10].

Overall, these cases demonstrate that combining FinOps best practices, cost visibility, and automation helps reduce waste and improve financial control in cloud environments.

6 Conclusion

It is undeniable that cloud computing has transformed the IT industry. By providing reliable, scalable, and on-demand services, cloud providers have established themselves as a fundamental component of modern IT organizations. However, the same characteristics that make cloud services so valuable also create significant challenges in terms of financial management. As discussed throughout this paper, unnecessary cloud costs can originate from various sources, including idle resources, unused storage, and inefficient data management practices.

To address these challenges, FinOps plays a vital role. It has emerged as an effective framework that helps organizations implement practices such as collaboration, visibility, shared responsibility, and continuous optimization. In addition to FinOps principles, organizations have adopted several widely recognized cost optimization techniques, including resource scheduling, the use of reserved and spot instances, storage optimization, and governance policies supported by resource tagging.

As cloud environments continue to grow in complexity, effective cost management will become increasingly important. While managing cloud costs was a much greater challenge in the past, organizations today have access to a wide range of tools, methodologies, and best practices that simplify this process. By combining FinOps principles with proven optimization techniques, organizations can maximize the value of cloud computing while maintaining financial control and operational efficiency.

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